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Age at menopause and duration of reproductive period in association with dementia and cognitive function: A systematic review and meta-analysis

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ABSTRACT

Introduction: The preponderance of dementia among postmenopausal women compared with same-age men and the female sex hormones neuroprotective properties support a tentative role of their deficiency in the dementia pathogenesis.
Methods: Pairs of independent reviewers screened 12,323 publications derived from a search strategy for MEDLINE to identify articles investigating the association of age at menopause/reproductive period with (i) dementia and (ii) cognitive function; a snowball of eligible articles and reviews was conducted and authors were contacted for additional information. Random-effect models were used for the meta-analysis.
Results: Age at menopause (13 studies; 19,449 participants) and reproductive period (4 studies; 9916 participants) in the highest categories were not associated with odds of dementia (effect size [ES]: 0.97 [0.78–1.21]) and Alzheimer's disease (ES: 1.06 [0.71–1.58]). Significant heterogeneity was however noted in both analyses (I^2: 63.3%, $p = 0.003$ and I^2: 72.6%, $p = 0.01$, respectively). Subgroup analyses by outcome assessment, study design, level of adjustment and study quality did not materially change the findings. In 9/13 studies assessing cognitive function, advanced age at menopause/longer reproductive period was significantly associated with better cognitive performance/lower decline. Due to statistical differences, no meta-analysis was possible for cognitive function.
Conclusions: Existing evidence does not support an association between indices of prolonged exposure to female hormones and lower dementia risk. There are indications, however, for better cognitive performance and delayed cognitive decline, supporting a link between female hormone deficiency and cognitive aging. Current literature limitations, indicated by the heterogeneous study-set, point towards research priorities in this clinically relevant area.

1. Introduction

During the menopausal transition, hormonal changes, involving mainly sex steroids, occur (Butler and Santoro, 2011). Consequently, the nervous system, closely regulated by steroid hormones, undergoes a sequence of biochemical, structural and functional changes (Sellers et al., 2015). There is evidence in support of a role of female hormone deprivation following menopause in the pathogenesis of dementia. Indeed, a higher incidence of dementia, especially Alzheimer's disease (AD), is observed in elderly

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women compared to men, amplified with increasing age, which cannot be explained by increased longevity (Andersen et al., 1999). Furthermore, animal studies demonstrate that estrogens and progestogenes exert various neuroprotective properties (Pike et al., 2009).

Many trials have investigated the effect of hormone therapy (HT) on cognition, using different combinations of estrogens and progestogenes. To date, findings do not support HT recommendation for prevention or treatment of postmenopausal cognitive impairment; it should be noted though, that studies have mainly focused on the traditionally used synthetic conjugated equate estrogens (CEE) and medroxyprogesterone acetate (O'Brien et al., 2014), whereas the scarce data on the human hormones 17β-estradiol and natural progesterone seem more optimistic (Boccardi et al., 2006; Gleason et al., 2005; Lord et al., 2008; Nilsen and Brinton, 2002; Robertson et al., 2009; Singh and Su, 2013). Nevertheless, exogenous hormone administration may exert a different action on the brain compared to endogenous exposure. Physiologically, during reproductive years, the brain is exposed to continuous estradiol fluctuations across the menstrual cycle, and intermittently after ovulation to progesterone or its neurosteroid metabolites (Guennoun et al., 2015). Pharmaceutical agents include synthetic progestins that differ from naturally produced progesterone in their central nervous system effects (Singh and Su, 2013). Interestingly, studies on serum concentrations of estrone and estradiol in pre- and post-menopausal women do not show any consistent association with cognitive performance (Henderson and Popat, 2011); yet, individual hormone levels sensitivity may account, as estrogen receptors polymorphisms have been associated with cognition (Sundermann et al., 2010).

Age at menopause and reproductive period duration, defined as the interval between menarche and menopause, comprise indices of the lifetime exposure to female sex hormones. These indices have been associated with postmenopausal diseases, including cardiovascular disease, but also depression as was recently demonstrated by our group (Atsma et al., 2006; Georgakis et al., 2016). To date though, the findings on the association of age at menopause with dementia and cognitive function remain inconclusive (Geerlings et al., 2001; Ryan et al., 2014; Zucchella et al., 2012). Therefore, we aimed to systematically review existing literature and synthesize the evidence assessing whether a later menopause or a longer reproductive period is associated with lower risk of dementia or better cognitive performance in postmenopausal women.

2. Methods

This systematic review is based on a pre-defined protocol following the MOOSE statement guidelines (Supplementary Table 1) (Stroup et al., 2000).

2.1. Search strategy

A systematic literature review was performed to identify relevant scientific articles. Medline was searched from inception to January 1st, 2015, using the MeSH terms "climacteric", "menopause" combined with "dementia", "Alzheimer's disease" and "cognitive impairment." No language or publication year restrictions were set. Reference lists of the included studies, and relevant reviews, were hand-searched for additional eligible studies ("snowball procedure"). All studies were evaluated for possible overlap in the included populations. Authors of studies not directly meeting eligibility criteria, but availing necessary data were contacted and asked to provide missing information, additional analyses as appropriately, or subject level data. If multiple publications on the same cohort were identified, the most recent publication or the one with the largest sample was selected, but information from all other relevant studies was retained. The detailed search strategy is available as supplementary material.

2.2. Exposure and outcome definition

Age at menopause was primarily defined as the age 1 year following last menstruation (World Health Organization, 1996 http://www.who.int/iris/handle/10665/41841#sthash.Dj1VDmHP.dpuf.

Published 1996. Accessed February 10, 2016), but other definitions of individual studies (i.e. age at final menstruation) were considered according to the study criteria. Reproductive period was determined as age at menopause minus age at menarche.

Two primary outcomes were tested; dementia and cognitive function. Assessment of dementia should have been based on clinical diagnostic criteria or neuropsychological tests defining severe cognitive impairment, which were validated for detection of dementia. AD was a secondary outcome. Regarding cognitive function, the use of validated neuropsychological tests assessing global cognition or specific cognitive domains was required; the individual tests used by eligible studies were classified to the respective domains.

2.3. Eligibility criteria

Eligible were all case-control, cross-sectional and longitudinal cohort studies investigating the association of age at menopause/reproductive period with dementia/cognitive function in postmenopausal women. Excluded were case series, case reports, in vitro, and animal studies. Studies encompassing solely women with dementia or pre-existing diseases with cognitive deficits (e.g. Down syndrome) were excluded. Cohorts including only perimenopausal participants and breast cancer survivors were deemed ineligible, as were studies exclusively concerning women with surgical menopause to avoid operative indications bias.

2.4. Data extraction and quality assessment

The data were abstracted in a pre-designed spreadsheet and included publication details, study/participants characteristics, and statistical analysis results. Maximally adjusted effect estimates were preferred. If the above data were not presented, missing information was sought in previous publications on the same population, the authors were contacted or crude effect estimates were calculated from available data. Eligible studies were evaluated on quality using Newcastle-Ottawa Scale (NOS) (Wells et al., 2011), as detailed in the supplementary material.

Six reviewers, in pairs of 2, performed independently and masked to each other the study selection, data abstraction and quality evaluation; disagreements were resolved by consensus.

2.5. Statistical analysis

2.5.1. Dementia

The effect estimates, namely Odds Ratios (OR) for cross-sectional and case-control studies, Hazard Ratios (HR) and Rate Ratios (RR) for longitudinal studies, along with their 95%CIs, on the risk of dementia were combined and pooled estimates were calculated using random-effect models (DerSimonian and Laird, 1986). Regarding studies providing raw data, multivariate analyses were performed (detailed in supplementary material) and individual study effect estimates were subsequently included in the data synthesis. A highest versus lowest exposure category approach was decided as this maximized the synthesized evidence of published studies. Between-study heterogeneity was estimated by Cochran

Q (significance at $p < 0.10$) and I^2 statistics. For the overall effect statistical significance level was set at $p < 0.05$.

Analyses were initially performed by synthesizing studies on all-cause dementia and AD. A subgroup analysis by dementia assessment (clinical diagnosis of dementia, screening neuropsychological test for severe cognitive impairment, death certificate report) was performed. All analyses were stratified by study design in order to restrict heterogeneity by different effect estimates (Deeks et al., 2008). Sensitivity analyses were performed by adjustment level and menopause type (only natural, both natural and artificial) and subgroup analysis by exposure reference category. Meta-regression analyses were conducted to investigate whether age, proportion of surgical menopause and HT use, adjustment level, study design and exposure categorization altered the outcome. Publication bias in analyses including ≥10 study arms (Siristatidis et al., 2013) was assessed by Egger's test (significance level at $p < 0.10$) (Egger et al., 1997). STATA Software, version 11.1 (STATA Corporation, College Station, TX, USA) was used for statistical analysis.

2.5.2. Cognitive function

Due to between-study heterogeneity regarding statistical methodology, no meta-analysis could be conducted for cognitive function.

3. Results

3.1. Search strategy results

Fig. 1 depicts the study selection process. The literature search yielded a total of 12,323 records. The full-texts of 133 potentially eligible articles, along with additional 14 identified through snowball, were evaluated for eligibility. Of these, 25 were deemed eligible (Amaducci et al., 1986; Baldereschi et al., 1998; Bove et al., 2014; Broe et al., 1990; Geerlings et al., 2001; Henderson et al., 2003; Hesson, 2012; Heys et al., 2011; Hong et al., 2001; Lebrun et al., 2005; Low et al., 2005; McLay et al., 2003; Mitchell et al., 2003; Nappi et al., 1999; Paganini-Hill and Henderson, 1996; Rasgon et al., 2005; Recchia et al., 2014; Roberts et al., 2006; Ryan et al., 2009, 2014; Smith et al., 1999; Tierney et al., 2013; Waring et al., 1999; Whalley et al., 2004; Zucchella et al., 2012). Fourteen concerned the association of age at menopause/reproductive period with dementia and 13 with cognitive function.

3.2. Dementia outcome

3.2.1. Description of studies

Table 1 summarizes the 14 studies assessing dementia characteristics. Seven case-control (#1-7, numbered in Table 1), 2 cross-sectional (#8-9), and 5 longitudinal cohort studies (#10-14) were included. The case-control studies encompassed 1135 dementia cases and 2434 controls, whereas the 7 cross-sectional and longitudinal studies included 16,324 postmenopausal women, among whom 2125 dementia cases. Eleven studies defined dementia/AD by clinical diagnostic criteria (#1–2, 4–8, 10, 12–14), 1 by death certificates (#3) and 2 assessed severe cognitive impairment using neuropsychological tests (#9, 11). The participants age ranged from 41 to 100 years. Nine studies provided information on type of menopause (#1, 3–4, 6–7, 10–13), but only in 2 we were able to obtain and reanalyze the data concerning women with exclusively natural menopause (#3, 13). All eligible studies included women who were current or past HT users; the ever HT users proportion varied from 0.9% (#5) to 45.8% (#11). All studies controlled for age, except for 1 (#7), but 5 did not adjust the analyses for other confounding factors (#1–2, 4–6). Besides age, most commonly used adjustment variables were education (6 studies), smoking (5 studies), HT use (5 studies), body mass index (BMI; 5 studies), number of children/parity (4 studies), depression (2 studies) and Apolipoprotein E (APOE) genotype (2 studies).

3.2.2. Quality assessment

Supplementary Table 3A presents NOS scoring for eligible studies. Generally, eligible studies scored high, with only 1 losing >2 points (#7). Almost all studies received the maximum points for sample selection. Case-control studies were compromised by poor comparability in factors other than age and high rates of non-respondents, whereas the inadequacy of follow-up cohorts was a quality limitation of cohort studies.

3.2.3. Data synthesis

Apart from 1 study (#4), presenting a continuous analysis for age at menopause, the remaining 13 were quantitatively synthesized (#1–3, 5–14). All studies provided data on age at menopause, and 4 on reproductive period (#6, 9, 10, 13). Regarding age at menopause, no statistically significant results were yielded from the synthesis for dementia (13 studies; 19,449 women; Fig. 2A) and AD (9 studies; 8626 women; Supplementary Fig. 1).

In sub-analyses addressing the significant heterogeneity (Table 2), a significant inverse association of age at menopause with dementia risk was found when the 2 cross-sectional studies (#8–9) were synthesized; a marginal inverse association also emerged from the 2 studies (#12–13) controlling for depression. Even the highest quality studies (#3, 6, 8–10, 12–13) and the 2 studies (#3, 13) referring exclusively to women experiencing a natural menopausal transition were heterogeneous. Studies comparing extreme age at menopause categories (>5-year difference) showed an inverse association between age at menopause and dementia, whereas studies treating age at menopause as dichotomous (low as reference) showed a harmful effect of late menopause; this subgroup analysis resolved heterogeneity. Fig. 3A presents the association of age at menopause with dementia and Alzheimer's disease, by method of assessment. When dementia was defined, as severe cognitive impairment, by neuropsychological tests (#9, 11) an inverse association with age at menopause emerged, whereas the meta-analyses of studies on clinically defined dementia and AD were also characterized by heterogeneity.

Likewise, no significant association was documented between duration of reproductive period and dementia (4 studies; 9916 women; Fig. 2B) or AD (3 studies; Supplementary Fig. 2) risk with significant heterogeneity in both analyses. Due to the paucity of studies reporting estimates for reproductive period, the respective sub-analyses were not conducted. The sub-analyses by method of dementia diagnosis were also compromised by the paucity of relevant studies (Fig. 3B).

3.2.4. Meta-regression analysis and publication bias

The meta-regression analysis results are presented in Supplementary Table 4. Mean age, HT use and proportion of surgically menopausal women did not significantly impact the outcome of interest. However, increasing difference between the highest and lowest age at menopause categories strengthened the inverse association between age at menopause and dementia ($p = 0.05$). Considering the paucity of available studies, meta-regression analysis should be interpreted with caution. In the main analysis of age at menopause and dementia, no publication bias was present (13 studies; Egger's $p = 0.92$).

Table 1
Characteristics of the included case-control (a) and cross-sectional/longitudinal cohort (b) studies for the risk of dementia.

A) Case-control studies

Study (Author, Year)	Place (study period)	Case characteristics	Control recruitment	Cases (N)	Controls (N)	Matching factors	Assessment of outcome	Age range (years)	Exposure variable (definition and assessment)	HT use	Type of menopause	Adjustment factors
#1 Amaducci et al. (1986)	Italy (1982–1984)	Consecutive AD patients admitted to neurology departments with a first-degree relative available for interview	(i) Hospital controls: patients admitted to other departments of the hospital where the cases were recruited. (ii) Population controls: neighbors, friends, acquaintances of cases. Excluded were individuals with dementia, according to Blessed Dementia Scale, or individuals related to a case	62	115	Age (±3 yrs), gender, region of residence	AD: Diagnosed based on a standardized protocol of the authors including neuropsychological, laboratory and clinical examination	41–80	AM Interview with next-of-kin of cases and controls by medically unsophisticated interviewers (retrospective assessment)	Cases: 10%, Hospital controls: 13.3%, Population controls: 8% (during menopause)	Natural and surgical	None
#2 Broe et al. (1990)	Sydney, Australia (1986–1989)	Demented subjects identified by general practitioners, referred to two hospitals and diagnosed with AD, capable of speaking English and with a suitable informant.	Up to 4 subjects for each case were identified by general practitioners and approached by phone. Excluded were those with MMSE score <26, unable to be examined in English or without a suitable informant.	106	106	Age (±2 yrs), gender	Diagnosis of possible and probable AD by NINCDS-ADRDA criteria.	cases: 52–93, controls: 53–96	AM Home interviews with informants by trained staff naïve to case-control status and true purpose of study (retrospective assessment)	Cases: 13.2%, Controls: 17% (current users)	NR	None
#3 Paganini-Hill and Henderson (1996)	California, USA (1981–1995)	Female deceased participants of "Leisure World cohort" with dementia diagnosis on death certificate without recorded multi-infarct dementia or dementia associated with another likely cause.	Subjects selected from the female cohort participants who had died in the period 1981–1995 without known dementia	148 (all-cause dementia), 74 (AD)	782	Year of birth (±1 yr), year of death (±1 yr)	Diagnosis of dementia, senile dementia, senility and AD as derived by death certificates.	72–104	AM (age at final menstruation) Mailed self-administered questionnaire (retrospective assessment)	Cases: 32.1%, Controls: 44.6% (ever users)	Natural	Marital status, BMI, parity, smoking, estrogen use
#4 Waring et al. (1999)	Rochester, Minnesota, USA (1980–1984)	Female local residents identified through the "Rochester Epidemiology Project", a diagnostic indexing and medical records linkage system, at the onset of AD.	Female local residents identified through the "Rochester Epidemiology Project" with sufficient medical evaluation to exclude dementia before and during the index year.	222	222	age (±3 years), approximate length of enrollment in the records-linkage system	AD diagnosis by DSM-III-R and NINCDS-ADRDA criteria	NR	AM Interview by trained nurses naive to case-control status and study hypothesis (retrospective assessment)	Cases: 15%, Controls: 21% (ever users)	Natural and surgical	None
#5 Hong et al. (2001)	Beijing, China	Female participants of "Dementia Prevalence Survey" diagnosed with AD. Excluded were those in the > 50th percentile MMSE score, with Hachinski ischemic score index > 7, pseudo-dementia due to depression, serious mental illness and other cause dementia.	Women from the same cohort, with MMSE score > 50th percentile without known cognitive impairment severe cerebrovascular or other central nervous system disorders.	115	1041	Age	MMSE, Global Deterioration Scale questionnaires	≥55 years	AM Interview by trained staff (retrospective assessment)	Cases: 0.9%, Controls: 1.7%	NR	None

Table 1 (*Continued*)

A) Case-control studies

Study (Author, Year)	Place (study period)	Case characteristics	Control recruitment	Cases (N)	Controls (N)	Matching factors	Assessment of outcome	Age range (years)	Exposure variable (definition and assessment)	HT use	Type of menopause	Adjustment factors
#6 Roberts et al. (2006)	Rochester, MS, USA (1985–1989)	Local residents with AD identified through a medical records-linkage system ("Rochester Epidemiology Project"). Excluded were women with unknown type of menopause.	Women identified through the same records-linkage system, free of dementia, and residing in Rochester the year of onset of dementia in the matched case (index year)	207	207	Age (±1 year)	Diagnosis of AD by DSM-IV criteria	NR	AM (1 y following last menstruation) and RP (AM minus age at menarche) Abstracted from medical records (retrospective assessment)	Cases: 11.4%, Controls: 10.6% (ever users)	Natural and surgical	Type of menopause
#7 Zucchella et al. (2012)	Pavia, Varese; Italy (2007–2010)	Female postmenopausal outpatients with AD referred to an AD Unit or Neurology Department.	Female postmenopausal outpatients to the same hospitals for non-cognitive neurological complaints. Subjects with Parkinson's disease or cerebrovascular lesions were excluded.	275	276	None	Diagnosis of AD by NINCDS-ADRDA criteria.	Cases: 53–96, Controls: 52–91	AM Structured interview (retrospective assessment)	Cases: 2.2%, Controls: 11.6% (ever users)	Natural and surgical	None

B) Cross-sectional and longitudinal cohort studies

Study (Author, Year)	Region (study period)	Study design	Cohort characteristics	Assessment of outcome	Cohort size	Incident cases in cohort	Mean follow-up (years)	Age range (years)	Exposure variable (definition and assessment)	HT use	Type of menopause	Adjustment factors
#8 Baldereschi et al. (1998)	Italy (1992–1993)	Cross-sectional	Community-dwelling and institutionalized participants in the "ILSA cohort study" selected from population registries	Diagnosis of possible/probable AD by DSM-III-R and NINCDS-ADRDA criteria	1582	92	NA	65–84	AM Home interview with participants and proxies (retrospective assessment)	11.6% (ever users)	NR	Age, education, age at menarche, smoking, alcohol intake, body weight at the age of 50, number of children
#9 Rasgon et al. (2005)	Sweden (1998–2001)	Cross-sectional	Pairs of twin women identified through the "Swedish Twin Registry", aged <85 years, without missing data on HT and educational level. Excluded were individuals who used more than one HT categories or progestin only and with data obtained through informants	Scoring ≤13.5 in TELE were indicative of cognitive impairment	5509 for AM; 5829 for RP	911 for AM; 977 for RP	NA	65–84	AM and RP Telephone interview (retrospective assessment)	22.6% (ever users)	NR	Age, education, parity, postmenopausal BMI, and smoking
#10 Geerlings et al. (2001)	Ommourd, Rotterdam; Netherlands (1990–1999)	Longitudinal cohort	Female local residents ≥55 years free of dementia at baseline, and with complete gynecological history. Excluded were those with highly unlikely AM and those with menopause induced after quitting oral contraceptives.	Diagnosis of dementia and AD based on DSM-III-R and NINCDS-ADRDA criteria respectively	2736	167 (all-cause dementia), 135 (AD)	6.3	≥55	AM (1 y following last menstruation) and RP (AM minus age at menarche) Home interviews by trained research assistants (retrospective assessment)	10.5% (ever users)	Natural, surgical and chemical	Age, education, smoking, alcohol intake, BMI, use of HT, number of children, APOE genotype

Table 1 (*Continued*)

B) Cross-sectional and longitudinal cohort studies

Study (Author, Year)	Region (study period)	Study design	Cohort characteristics	Assessment of outcome	Cohort size	Incident cases in cohort	Mean follow-up (years)	Age range (years)	Exposure variable (definition and assessment)	HT use	Type of menopause	Adjustment factors
#11 Mitchell et al. (2003)	Beaver Dam, WI, USA (1988–2000)	Longitudinal cohort	Local residents aged 43–84 years at baseline (1988–1990), who participated in "Epidemiology of Hearing Loss Study" and were alive in 1993 follow-up examination.	Scoring <17 in 21-point MMSE or <24 in 30-point MMSE were indicative of cognitive impairment	1172	94	5.2	53–97	AM Interview by trained staff at baseline and each follow-up (retrospective assessment)	45.8% (ever users)	Natural and surgical	Age, education
#12 Ryan et al. (2014)	Montpellier, Bordeaux, Dijon; France (1999–2008)	Longitudinal cohort	Non-institutionalized individuals randomly recruited from electoral rolls, free of dementia at baseline without missing data concerning key covariates.	Diagnosis of dementia based on DSM-IV-R criteria	3739	393	~7	≥65	AM (1 y following last menstruation) Interview (retrospective assessment)	20.3% (users at menopause)	Natural, surgical and chemical	Age, education, baseline cognitive function, recruitment centre physical limitations, chronic illness, depression, use of HT at the menopause and current HT use, APOE4 genotype
#13 Bove et al. (2014)	USA (1993–2014)	Longitudinal cohort	MAP: Older women from Chicago free of known dementia at baseline. ROS: Older Catholic nuns free of known dementia at baseline. MARS: African American women aged ≥65 years, free of dementia at baseline.	Neurologic examination and diagnosis of dementia and AD based on NINCDS-ADRDA criteria	942 for AM; 937 for RP	Dementia 144, 123 CE	6.6	53–100	AM and RP Interview by trained staff (retrospective assessment)	33.6% (ever users)	Natural	Age, marital status, HT use, depression, race, BMI, smoking status
#14 Recchia et al. (2014)	Varese, Italy	Longitudinal cohort	Residents of Varese province aged ≥80 years registered through municipality registry.	Diagnosis of dementia based on DSM-IV criteria	644	293	3.8	≥ 80	AM Obtained through a standardized questionnaire from the subject and primary informant (retrospective assessment)	8.2	Natural, surgical	Age, education

NA: Not applicable, NR: Not reported.

HT: Hormone therapy, AD: Alzheimer's disease, AM: Age at menopause, RP: Reproductive period, BMI: Body mass index, MMSE: Mini-Mental State Examination, NINCDS-ADRDA: National Institute of Neurological and Communicative Disorders and Stroke and the Alzheimer's Disease and Related Disorders Association, DSM: Diagnostic and Statistical Manual of Mental Disorders, APOE: Apolipoprotein E, ILSA: Italian Longitudinal Study on Aging, MAP: Memory and Aging Project, ROS: Religious Orders Study, MARS: Minority Aging Research Study.

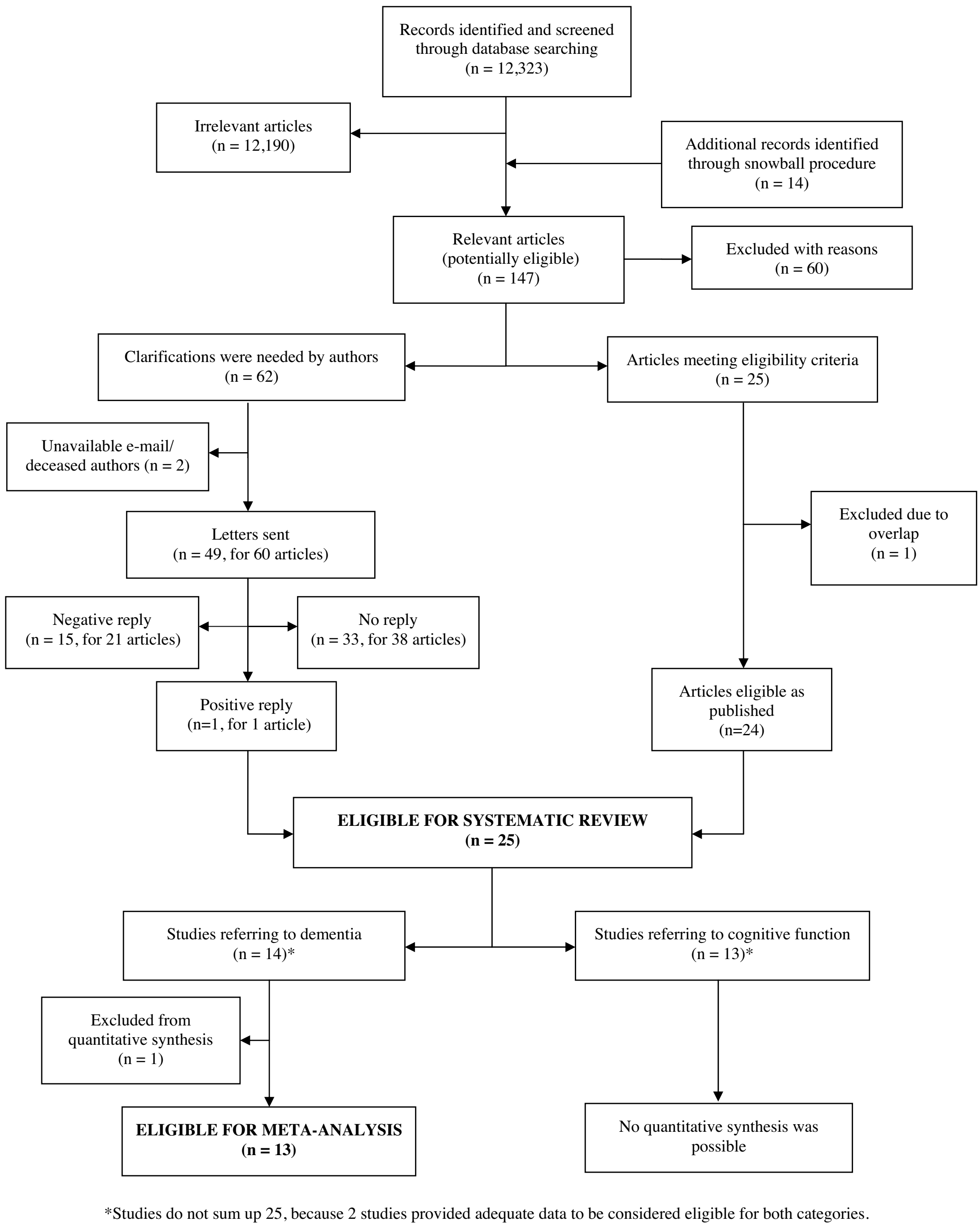

Fig. 1. Flow chart presenting the selection of the eligible studies.

3.3. *Association of exposures by cognitive function*

3.3.1. *Description of studies*

Table 3 summarizes the sample characteristics of the 8 cross-sectional (#15–22, as numbered in Table 3) and 5 longitudinal cohort studies (#13–14, 23–25), on cognitive function, encompassing a total of 19,328 women. Nine studies included solely women with natural menopause (#12–13, 15, 17–20, 22, 24), 5 excluded HT users (#15, 18, 20, 23–24), and 9 adjusted results for confounders (#12–13, 16, 18–20, 22–23, 25) including age, HT use, marital status, education, BMI and depression.

3.3.2. *Assessment of quality*

Only 2 cross-sectional studies lost ≥2 quality points (#15, 21) (Supplementary Table 3B). Main limitations included self-administered questionnaires for exposure ascertainment, and poor comparability between exposed and unexposed samples.

3.3.3. *Summary results on single cognitive measurements*

The by necessity descriptive results of the studies on cognitive function are presented in Table 4. Twelve studies (#12–13, 15–22, 24–25) presented associations of age at menopause/reproductive period with cognitive function assessed at a single time-point. Advanced age at menopause was statistically significantly asso-

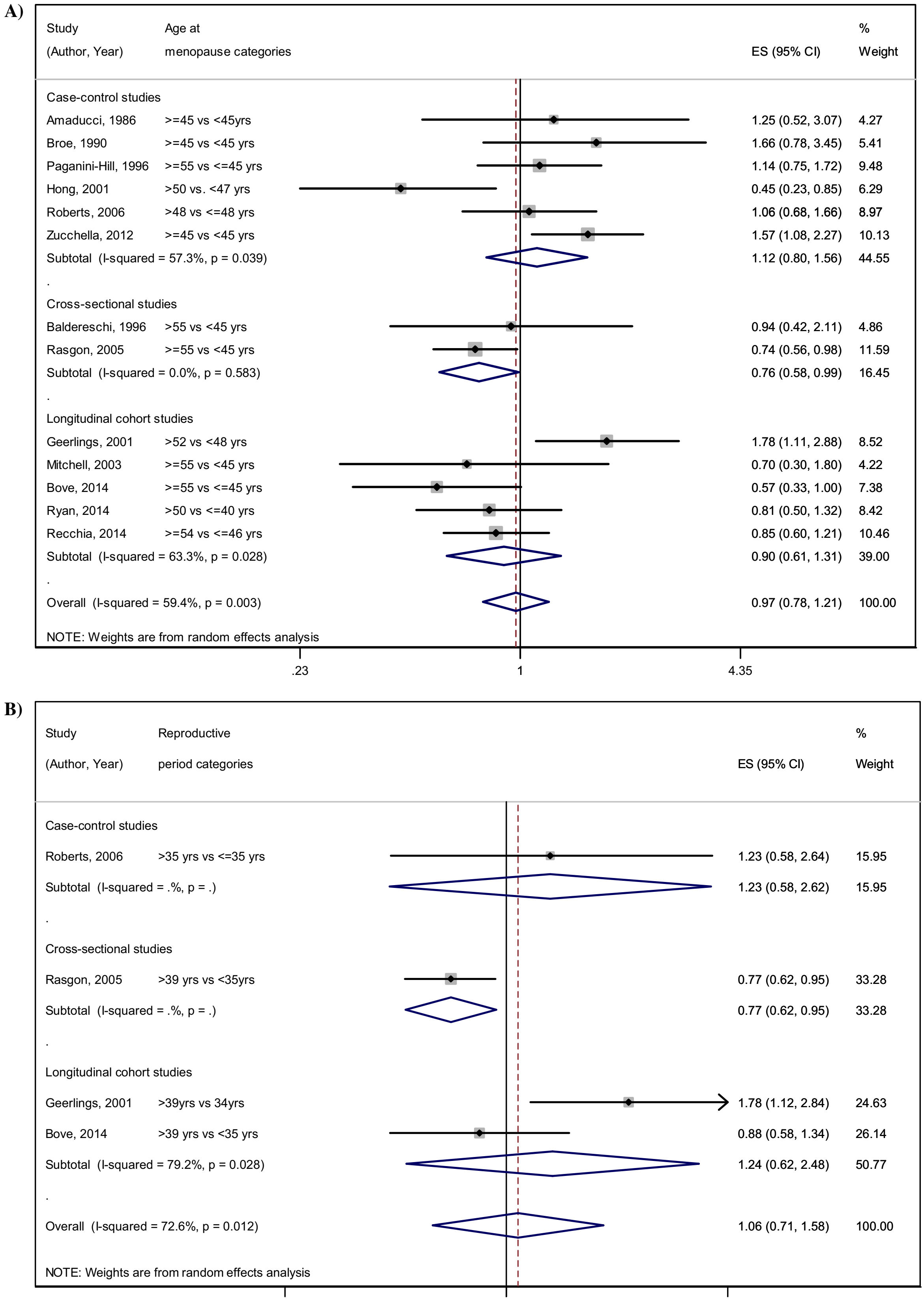

Fig. 2. Association between (A) age at menopause and (B) duration of reproductive period with all-cause dementia by study design.
Thirteen studies reported the association of age at menopause (highest versus lowest category) with all-cause dementia in postmenopausal women ($P = 0.79$), whereas 4 studies were included in the reproductive period analysis ($P = 0.79$). Effect sizes (ES) in the individual studies are indicated by the data markers (shaded boxes around the data markers reflect the statistical weight of the study); 95% Confidence Intervals (CI) are indicated by the error bars. The pooled-effect estimate with its 95% CIs is depicted as a diamond. Subgroup analyses by study design are also presented

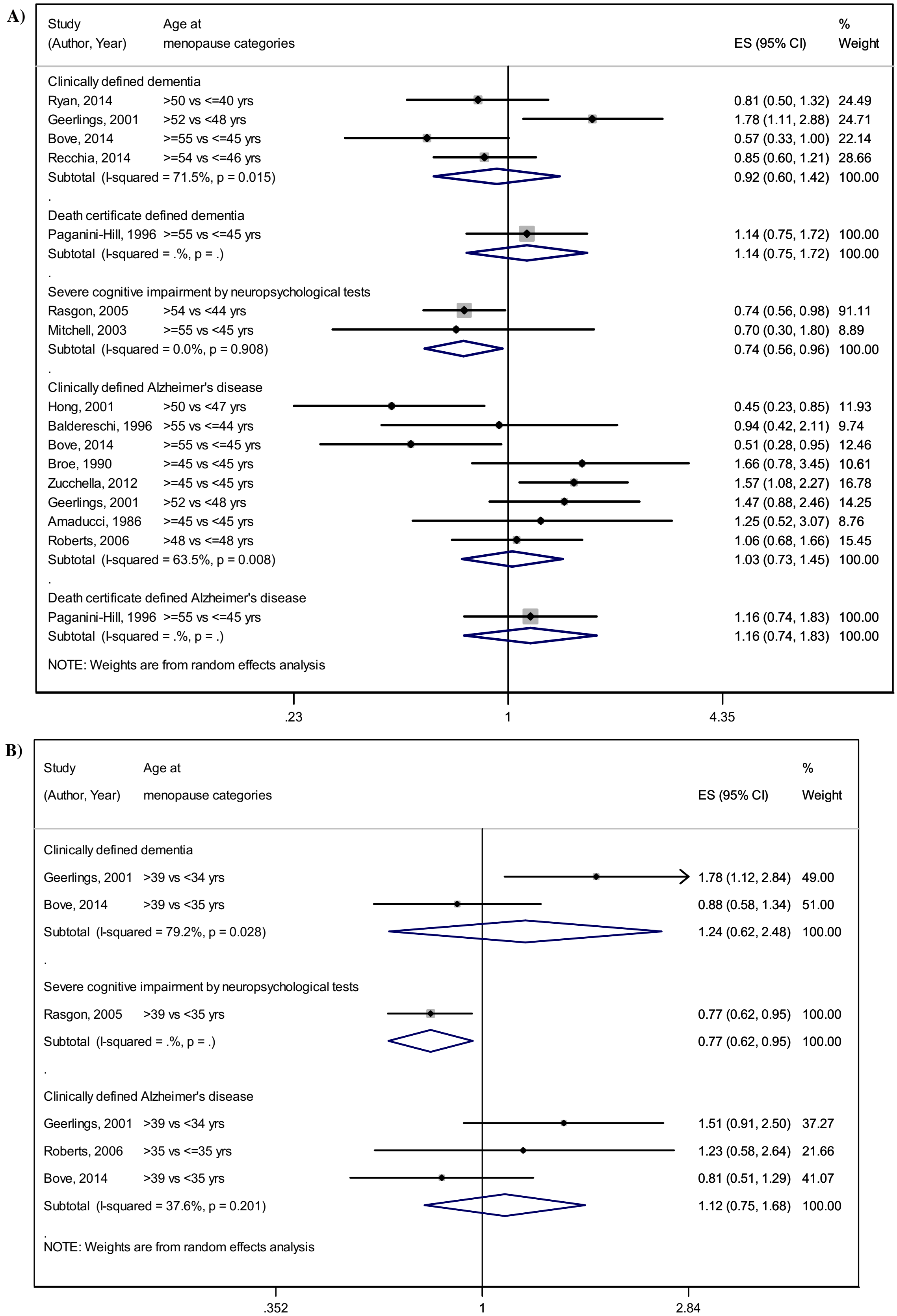

Fig. 3. Association of (A) age at menopause and (B) duration of reproductive period with dementia and Alzheimer's disease by method of diagnosis. Effect sizes (ES) in the individual studies are indicated by the data markers (shaded boxes around the data markers reflect the statistical weight of the study); 95% Confidence Intervals (CI) are indicated by the error bars. The pooled-effect subgroup estimates with their 95% CIs are depicted as diamonds.

Table 2
Summary effect sizes (ES) and 95% Confidence Intervals (CI) of (i) age at menopause and (ii) duration of the reproductive period and risk of all-cause dementia and Alzheimer's disease by study design, method of dementia assessment, level of adjustment, type of menopause exposure categorization and study quality. The effect estimates refer to analyses for the highest versus lowest category of age at menopause and duration of the reproductive period, as provided by the individual studies.

Analyses	All-cause dementia			Alzheimer's disease		
	k[*]	ES (95% CI)	Heterogeneity (I^2, p)	*k*[*]	ES (95% CI)	Heterogeneity (I^2, p)
Age at menopause						
Overall analysis	13	0.97 (0.78–1.21)	63.3%, 0.003	9	1.05 (0.78–1.41)	58.4%, 0.01
Study design						
Longitudinal cohort studies	5	0.90 (0.61–1.31)	63.3%, 0.03	2	0.88 (1.31–2.47)	84.8%, 0.01
Cross-sectional studies	**2**	**0.76 (0.58–0.99)**	**0.0%, 0.58**	1	0.94 (0.42–2.11)	–
Case-control studies	6	1.12 (0.80–1.56)	57.3%, 0.04	6	1.14 (0.85–1.53)	41.8%, 0.10
Outcome assessment						
Clinical diagnosis of dementia	5	0.97 (0.69–1.36)	64.4%, 0.02	–	–	–
Clinical diagnosis of AD[§]	–	–	–	8	1.03 (0.73–1.45)	63.5%, 0.008
Neuropsychological screening	**2**	**0.74 (0.56–0.96)**	**0.0%, 0.91**	–	–	–
Levels of adjustment						
Adjusted effect estimates[†]	8	0.90 (0.71–1.14)	50.0%, 0.05	4	0.98 (0.63–1.53)	57.3%, 0.07
Age	11	0.91 (0.73–1.13)	53.5%, 0.02	7	0.96 (0.68–1.35)	59.0%, 0.02
Education	6	0.92 (0.70–1.2)	51.3%, 0.07	2	1.29 (0.84–1.99)	0.0%, 0.36
Hormone Therapy	5	1.00 (0.68–1.46)	61.7%, 0.03	4	0.98 (0.63–1.53)	57.3%, 0.07
BMI[§]	5	0.96 (0.65–1.41)	70.2%, 0.009	4	0.98 (0.63–1.53)	57.3%, 0.07
Smoking	5	0.96 (0.66–1.41)	70.2%, 0.009	4	0.98 (0.63–1.53)	57.3%, 0.07
Parity	4	1.08 (0.71–1.64)	71.1%, 0.02	3	1.23 (0.90–1.68)	0.0%, 0.62
Depression	2	0.70 (0.49–1.01)	0.0%, 0.36	**1**	**0.51 (0.29–0.95)**	–
APOE genotype[§]	2	1.20 (0.56–2.60)	80.4%, 0.02	1	1.47 (0.88–2.46)	–
Type of menopause						
Solely naturally menopausal women	2	0.83 (0.43–1.62)	73.1%, 0.05	2	0.80 (0.36–1.77)	77.2%, 0.04
Age at menopause classification						
>5 years of difference between highest and lowest category	**7**	**0.81 (0.69–0.96)**	**0.0%, 0.58**	3	0.84 (0.50–1.42)	54.6%, 0.11
≤ 5 years of difference between highest and lowest category	2	0.91 (0.24–3.51)	91%, 0.001	2	0.83 (0.26–2.64)	87.2%, 0.005
No difference between highest and lowest category (dichotomous analysis)	**4**	**1.36 (1.06–1.76)**	**0.0%, 0.56**	**4**	**1.36 (1.06–1.76)**	**0.0%, 0.56**
Quality of studies[‡]						
High scoring studies	7	0.95 (0.73–1.24)	57.4%, 0.03	5	1.01 (0.73–1.40)	43.2%, 0.13
Reproductive period duration						
Overall analysis	4	1.06 (0.71–1.58)	72.6%, 0.01	3	1.12 (0.75–1.68)	36.6%, 0.20
Study design						
Longitudinal cohort studies	2	1.24 (0.62–2.48)	79.2%, 0.03	2	1.10 (0.60–2.01)	67.8%, 0.08
Cross-sectional studies	**1**	**0.77 (0.62–0.95)**	–	–	–	–
Case-control studies	1	1.23 (0.58–2.62)	–	1	1.23 (0.58–2.62)	–

§ AD: Alzheimer's disease, BMI: Body Mass Index, APOE: Apolipoprotein E. Bold indicates statistical significance ($p <= 0.05$).
* Number of study arms.
† Adjusted were considered effect estimates from studies using at least age, as well as another confounder, as adjustment or matching factor.
‡ Studies classified as high scoring in terms of quality were those that lost <2 points in the Newcastle-Ottawa Scale.

ciated with better global cognition (#18), working memory (#16, 24), non-verbal reasoning (#24), visual memory (#12) and verbal fluency (#12) in 4 out of 7 (#12–13, 15–16, 18, 21, 24) studies. Increasing reproductive period duration was related to higher cognitive performance in 4 (#12-13, 20, 22) out of 6 (#12–13, 17, 19–20, 22) studies; the specific domains included higher global cognition (#13, 20), verbal memory (#20, 22), visual memory (#22), working memory (#22) and verbal fluency (#25). Interestingly, the largest study, comprising 11,094 women (#20), found that longer reproductive period was significantly associated with enhanced global function and verbal memory.

3.3.4. Summary results on cognitive decline

Three prospective studies (#12, 23, 25) demonstrated significant inverse associations between later age at menopause and decline of global cognitive ability (#12, 23), psychomotor speed (#12) and executive function (#25) (Table 4; lowest panel). One study also found that longer reproductive period was related with lower executive function decline (#25).

It is noteworthy that no studies reported significant opposing effects; yet, the majority of them reported significant associations only in some of the cognitive domains analyzed; lastly no consistent association was present with a single domain in the totality of eligible studies.

4. Discussion

To date, the heterogeneous findings of published studies do not support a consistent association between early menopause or shorter reproductive period and dementia risk. Main limitations in their synthesis are related to heterogeneity of study designs and methodologies and complexities in assessing the time-varying outcome. Statistically significant associations of individual studies, though suggest that older age at menopause or more extended reproductive period may be associated with higher cognitive performance and delayed cognitive decline.

Therefore, the role of confounding factors should be considered in interpreting the results of our synthesis. Firstly, increasing

Table 3
Characteristics of eligible studies examining cognitive function.

Study (Author, Year)	Region (Study period)	Study design	Cohort characteristics	Assessment of outcome	Cohort size	Follow-up duration (years)	Age (range/mean ± SD)	Exposure variable (definition and assessment)	HT use	Type of menopause	Adjustment factors
#15 Nappi et al. (1999)	Pavia, Italy	Cross-sectional	Consecutive patients of a reproductive endocrinology unit for treatment of menopausal complaints	• Serial Learning Test	76	NA	52.4 ± 4.3 y	AM (1 y following last menstruation) Interview in the context of clinical evaluation (retrospective assessment)	0%	Natural	None
#16 Smith et al. (1999)	Southern California, USA	Cross-sectional	Subjects recruited through community outreach efforts and enrolled as comparison subjects in a longitudinal study of aging and dementia	• MMSE and OMC • CERAD-10 word recall and BNT • Animal fluency test and Controlled Oral Word Association Test • Judgment of Line Orientation and CERAD figure drawing • Digit Forwards and Digit Backwards from Wechsler • Visual Span Forward and Visual Span Backwards	87	NA	52–97 y	AM (retrospective assessment)	57%	NR	Age, education, number of previous tests, depressive symptoms
#17 Henderson et al. (2003)	Melbourne, Australia	Cross-sectional	Subjects recruited when aged 45–55 y through random-digit telephone dialing of local residents. Excluded were women with surgical menopause.	• A word list recall	139	NA	52–63 y	RP (AM, as 1 y following last menstruation, minus age at menarche) Abstracted by self-completed menstrual calendars (prospective assessment)	23%	Natural	None
#18 Lebrun et al. (2005)	Utrecht, the Netherlands (1993–1997)	Cross-sectional	Participants were recruited from a study enrolling healthy women aged 49–70 y who came for breast cancer screening. Inclusion criteria were a natural menopause 8–30 y ago, no reporting of HT use after menopause, an intact uterus with at least one intact ovary and capability of visiting the health centre.	• MMSE	402	NA	49–70 y	AM Abstracted by a cohort's baseline data	0%	Natural	Age, mean arterial pressure, BMI, education

Table 3 (*Continued*)

Study (Author, Year)	Region (Study period)	Study design	Cohort characteristics	Assessment of outcome	Cohort size	Follow-up duration (years)	Age (range/mean ± SD)	Exposure variable (definition and assessment)	HT use	Type of menopause	Adjustment factors
#19 Low et al. (2005)	Australia (2001)	Cross-sectional	Participants aged 60–64 y derived from a longitudinal cohort on depression, anxiety and cognition who were selected from the electoral rolls and agreed to participate	• MMSE • CVLT-List A • Digit Backwards from Weschler • Symbol-Digit Modalities Test • A test described by authors for reaction time	580	NA	60–64 y	RP (AM, as 1 y following last menstruation, minus age at menarche) Interview (retrospective assessment)	31%	Natural	Age, education, physical health, depression, verbal intelligence, BMI, diabetes, smoking, exercise level, alcohol, use of oral contraceptives, use of HT, number of children
#20 Heys et al. (2011)	Guangzhu, China (2005–2008)	Cross-sectional	Participants were recruited from a local community social and welfare association for the elderly with a paid (4 Yen) membership. Exclusion criteria: nulliparity, premenopausal status, past hysterectomy or bilateral oophorectomy, past, caurrent or unknown HT use, previous or current use of OCP, missing data for cognitive outcomes and biologically implausible reproductive milestones.	• MMSE • CER10 CE-word recall	11,094	NA	50–95 y	RP (AM minus age at menarche) Interview (retrospective assessment)	0%	Natural	Age. education, parental possessions, occupation and physical activity
#21 Hesson (2012)13	Victoria, Canada	Cross-sectional	Volunteers responding to advertisements for participation in a memory study in the elderly. They were considered for eligibility via telephone interview if they were self-considered in good physical and mental health and were able to provide information for the timing and nature of menopause and HRT use.	• A task described by authors for prospective memory	50	NA	69.3+-3.3 y	AM Interview (retrospective assessment)	NR	Narural (94%) and surgical (6%)	None

Table 3 (*Continued*)

Study (Author, Year)	Region (Study period)	Study design	Cohort characteristics	Assessment of outcome	Cohort size	Follow-up duration (years)	Age (range/mean ± SD)	Exposure variable (definition and assessment)	HT use	Type of menopause	Adjustment factors
#22 Tierney et al. (2013)	Toronto, Canada	Cross-sectional analysis of an RCT	Postmenopausal women recruited by primary care physicians or posters. Included subjects had memory complaints, scored below normal for their age on RAVLT, were free of diseases that might affect cognition or predispose to death within 2 y, unstable diseases the previous year, and conditions that might be exacerbated by estrogen, did not use HT the last 2 y, were fluent in English, had a natural menopause and were capable of recalling age at menopause.	• RCFT • CVLT	126	NA	60–89 y	RP (AM, as 1 y following last menstruation, minus age at menarche) Self-administered questionnaire (retrospective assessment)	25%	Natural	Age, education, number of children, breastfeeding, years since menopause, previous OC and HT use
#23 McLay et al. (2003)	East Baltimore, USA (1981–1996)	Longitudinal cohort	Participants selected from the general local population and recruited in a study investigating mental disorders. Excluded were women who were not postmenopausal, who used estrogens and who had complete MMSE data.	• MMSE	361	NA	31–94 y	AM Interview (retrospective assessment)	0%	Natural (55%) and surgical (44%)	Age, race, education, baseline cognitive function, childbirth
#24 Whalley et al. (2004)	Scotland, Great Britain (2001)	Longitudinal birth cohort	Women born in 1936 matched through local health registry and recruited by local family doctors.	• Raven's Progressive Matrices • Auditory Verbal Learning Test • Block design from Weschler • Digit Span from Weschler • Uses of Objects Test	144	NR	65 y	AM (1 y following last menstruation) Interview by trained staff (prospective, reassessment by telephone after 2 y)	0%	Natural	None

Table 3 (*Continued*)

Study (Author, Year)	Region (Study period)	Study design	Cohort characteristics	Assessment of outcome	Cohort size	Follow-up duration (years)	Age (range/mean ± SD)	Exposure variable (definition and assessment)	HT use	Type of menopause	Adjustment factors
#25 Ryan et al. (2009)	Montpellier, France (1999–2005)	Longitudinal cohort	Participants recruited randomly through local electoral rolls, aged ≥ 65 y and non-institutionalized who provided a written informed consent.	• MMSE • Isaacs Set Test • BVRT • TMA • TMB	996	4	≥65 y	AM (1 y following last menstruation) and RP (AM minus age at menarche) Obtained through questionnaire at clinical examination (retrospective assessment)	34%	Natural (81.3%) and surgical (18.7%)	Age, educational level, marital status, depressive symptoms, high caffeine intake, physical incapacities, comorbidity, age at menarche, age at first child birth, HT use
#12 Ryan et al. (2014)	Montpellier, Bordeaux, Dijon, France (1999–2001)	Longitudinal cohort	Non-institutionalized individuals randomly recruited from electoral rolls, free of dementia at baseline without missing data concerning key covariates.	• MMSE • Isaacs Set Test • BVRT • TMA • TMB	3739	7	≥65 y	AM (1 y following last menstruation) Interview (retrospective assessment)	21%	Natural	Recruitment centre, age, education, physical limitations, chronic illness, depression, HT use at menopause, current HT use
#13 Bove et al. (2014)	USA (1993–2014)	Longitudinal cohort	MAP: Older women from Chicago free of known dementia at baseline. ROS: Older Catholic nuns free of known dementia at baseline. MARS: African American women aged ≥ 65 y, free of dementia at baseline.	• Average z-score of 19 individual tests corresponding to different cognitive domains	1534	6.3	53–100 y	AM (age at final menstrual period) and RP (AM minus age at menarche) Interview by trained staff (retrospective assessment)	31%	Natural	Age, marital status, HT use, depression, race, BMI, smoking status

NA: Not applicable, NR: Not reported.

HT: Hormone therapy, AD: Alzheimer's disease, AM: Age at menopause, RP: Reproductive period, BMI: Body mass index, MMSE: Mini-Mental State Examination, OMC: Test of orientation, memory and concentration, CERAD: Consortium to Establish a Registry for Alzheimer's Disease, BNT: Boston Naming Test, RCFT: Rey Complex Figure Test, CVLT: California Verbal Learning Test, TMA: Trail Making A, TMB: Trail Making B, BVRT: Benton's visual retention test, MAP: Memory and Aging Project, ROS: Religious Orders Study, MARS: Minority Aging Research Study.

Table 4
Summary of results of studies exploring the association between age at menopause (AM) and/or reproductive period duration (RP) and global cognition or individual cognitive domains.

Study (Author, Year)	Country	Mean age	Type of menopause (1: Natural, 2: Surgical, 3: other)	HT use	Exposure	N	Direction of association between cognitive function and AM or RP									Adjustments
							Global cognition	*Verbal memory*	*Visual memory*	*Executive function/ Non-verbal reasoning*	*Working memory/ Non-verbal attention*	*Psychomotor speed/Reaction time*	*Verbal fluency*	*Visuo-spatial ability*	*Prospective memory*	
Associations with a single cognitive measurement																
Lebrun et al. (2005)	Netherlands	66.3	1: 100%	0%	AM	402	+**									Age, mean arterial pressure, BMI, education
Low et al. (2005)	Australia	62.5	1: 100%	31%	RP	580	ns	ns			ns	ns				Age, education, physical health, depression, verbal intelligence, BMI, diabetes, smoking, exercise level, alcohol, use of oral contraceptives, use of HT, number of children
Nappi et al. (1999)	Italy	52.4	1: 100%	0%	AM	76		ns								None
Smith et al. (1999)	USA	78.5	NR	57%	AM	87	ns	ns			+*		ns	ns		Age, education, number of previous tests, depressive symptoms
Whalley et al. (2004)	Scotland, Great Britain	65	1: 100%	0%	AM	144		ns		+**		+**		ns		None
Henderson et al. (2003)	Australia	56.7	1: 100%	23%	RP	139		ns								None
Heys et al. (2011)	China	60.4	1: 100%	0%	RP	11,094	+**	+**								Age. education, parental possessions, occupation and physical activity
Hesson (2012)	Canada	69.3	1: 94% 2: 6%	NR	AM	50									ns	None
Tierney et al. (2013)	Canada	74.1	1: 100%	25%	RP	126		+**	+*e		+*					Age, education, number of children, breastfeeding, years since menopause, previous OC and HT use
Ryan et al. (2009)	France	72.8	1: 81.3% 2: 18.7%	34%	RP	996	ns	ns	ns	ns		ns	+**			Age, educational level, marital status, depressive symptoms, high caffeine intake, physical incapacities, comorbidity, age at menarche, age at first child birth, HT use

Table 4 (*Continued*)

Study (Author, Year)	Country	Mean age	Type of menopause (1: Natural, 2: Surgical, 3: other)	HT use	Exposure	N	Direction of association between cognitive function and AM or RP									Adjustments
							Global cognition	*Verbal memory*	*Visual memory*	*Executive function/ Non-verbal reasoning*	*Working memory/ Non-verbal attention*	*Psychomotor speed/Reaction time*	*Verbal fluency*	*Visuo-spatial ability*	*Prospective memory*	
Ryan et al. (2014)	France	74.1	1: 100%	21%	AM	3842	ns		$+^{**}$	ns		ns	$+^{**}$			Recruitment centre, age, education, physical limitations, chronic illness, depression, HT use at menopause, current HT use
Bove et al.,	USA	77.9	1: 100%	31%	AM	1534	ns									Age, marital status, HT use, depression, race, BMI, smoking status
					RP	1528	$+^{***}$									Age, marital status, HT use, depression, race, BMI, smoking status
Associations with decline between two time-distinct cognitive measurements																
McLay et al. (2003)	USA	63.4	1: 55% 2: 45%	0%	AM	361	$+^{*}$									Age, race, education, baseline cognitive function, childbirth
Ryan et al. (2009)	France	72.8	1: 81.3% 2: 18.7%	34%	AM	996	ns	ns	ns	$+^{*}$		ns	ns			Age, educational level, baseline cognition
					RP	996	ns	ns	ns	$+^{*}$		ns	ns			
Ryan et al. (2014)	France	74.1	1:79% 2:10% 3:11%	21%	AM	3739	$+^{*}$		ns	ns		$+^{**}$	ns			Baseline cognition, recruitment centre, age, education, physical limitations, chronic illness, depression, HT use at menopause, current HT use

Study characteristics refer to the subpopulation included in the analysis presented in this table. If not available in the published manuscript characteristics of the total sample are presented.
The results with the highest level of adjustment are presented.
+: indicates significant association between increasing age at menopause and/or reproductive period duration with better cognitive function or lower cognitive decline.
ns: indicates non-significant association.
AM: Age at menopause, RP: Reproductive period duration, HT: Hormone Therapy, BMI: Body mass Index, OC: Oral contraceptives.

* $p<0.05$.

** $p<0.01$.

*** $p<0.001$.

age and associated comorbidities may overshadow the effect of cumulative sex hormone exposure on dementia risk. Considering that women with advanced age at menopause live longer (Jacobsen et al., 2003), and are consequently exposed to a higher dementia risk, it is possible that advanced age at menopause is overrepresented in late-life cohorts. To disentangle the effect of aging from the effect of sex hormones exposure, we conducted a sensitivity analysis of studies controlling for age and a meta-regression analysis for mean age of participants; none of them materially changed the findings. Available data did not allow stratification by age and use of age at onset of dementia as the outcome variable, which would have been more appropriate approaches. In 1 study examining the latter association, a delayed menopause was associated with a later AD onset (Corbo et al., 2011).

Secondly, comorbidities particularly depression and cardiovascular disease may alter findings, as they interfere with cognitive decline in postmenopausal women (de Bruijn and Ikram, 2014; Diniz et al., 2013). Their effect could not be statistically quantified in this meta-analysis as only few studies controlled for these factors. Age at menopause has been inversely associated with risk of cardiovascular disease (Atsma et al., 2006), whereas our group recently demonstrated that early menopause is a risk factor for depression in postmenopausal women (Georgakis et al., 2016).

Thirdly, both premature menopause and increased dementia risk represent different aspects of accelerated aging, and therefore common genetic predisposition may be present. APOE genotype, an established risk factor for AD (Farrer et al., 1997), has been reported to affect estrogen levels and natural menopause timing (Meng et al., 2012). The 2 studies adjusting for APOE genotype in the dementia meta-analysis had, however, conflicting findings (Geerlings et al., 2001; Ryan et al., 2014). On the other hand, reports from birth cohorts have shown that higher childhood cognitive ability is associated with a later menopause (Kuh et al., 2005; Whalley et al., 2004), a relationship possibly driven by the higher education and healthier lifestyles these women usually follow (Luoto et al., 1994). Smoking is another potential confounder, as it affects both timing of menopause and cognition (Anstey et al., 2007).

Other factors influencing the lifetime exposure of women to endogenous female hormones may impact on the findings. Age at menarche was not examined in this systematic review, as it is characterized by lower variability (80% of women experience menarche within a window of <3 years) (Chumlea et al., 2003), compared to age menopause (7-year window for 80% of women) (Gold et al., 2013) and therefore impacts to a lesser extent on duration of the reproductive period; nevertheless age at menarche is indirectly included in the analyses of reproductive period (defined as age at menopause minus age at menarche). Furthermore, we chose not to analyze age at menarche separately, as the longer interval between menarche and dementia onset makes their relationship subject to confounders. A few of the included studies have examined the direct effect of age at menarche on dementia risk and cognitive function and do not seem to support an association (Geerlings et al., 2001; Paganini-Hill and Henderson, 1996; Roberts et al., 2006; Ryan et al., 2009; Waring et al., 1999; Zucchella et al., 2012). Parity also affects exposure to endogenous estrogens, since multiparous women generally reach menopause later than the nulliparous, and there are indications from clinical and neuropathological studies that increasing number of children may be associated with increased risk of AD, without however consistent conclusions (Beeri et al., 2009; Colucci et al., 2006; Dunkin et al., 2005; Ptok et al., 2002; Sobow and Kloszewska, 2004); as only 4 of the studies adjusted their results for parity, its tentatively confounding effect cannot be precluded.

It is possible that only vast differences in the lifetime sex hormones exposure could outrange the intermediate confounders. Indeed, studies comparing women with extreme ages at menopause, showed significant associations between dementia risk and earlier menopause.

As opposed to studies assessing dementia, those reporting associations with cognitive function, were more consistently suggesting that older age at menopause or longer reproductive period was associated with better cognitive performance. Although the cross-sectional reports may be compromised by selection bias, studies assessing cognitive decline are more reliable given their prospective design. They are more likely however, to outline an association of lifetime hormone exposure with cognitive aging, rather than mild cognitive impairment and dementia; age-related cognitive decline has a differential neuropsychological profile and is not an inevitable precursor of dementia (Lindeboom and Weinstein, 2004). Notably, the lack of a consistent association with a particular domain among the studies raises concerns if the findings are robust or result from multiple comparisons.

Gonadal steroids may regulate cognition through different pathways inducing short- and long-term effects. For example, estradiol binds to nuclear receptors, which stimulate neurotransmitter metabolism, neurotrophins synthesis and synaptic formation in the hippocampus and prefrontal cortex. These effects have been associated with memory consolidation and improved cognitive performance in animal studies (Luine, 2014). In the same context, estradiol directly activates glutamate receptor and may protect against serotoninergic and cholinergic depletion, according to human studies (Greendale et al., 2011).

Given the inefficiency of up-to-date attempts to procure therapies by the estrogen deficiency hypothesis (Henderson, 2010; Resnick et al., 2009; Shumaker et al., 2003), other mechanisms could also apply for the female preponderance in dementia. An intriguing hypothesis is that sex differences in stress exposures and stress response partly mediate this phenomenon. Major stressful events have been associated with earlier age at AD onset and AD pathology in animals (Munro, 2014). Women seem to have a more abrupt increase in cortisol levels after stress exposure than men, which is even higher in older individuals (Otte et al., 2005). Furthermore, genetic polymorphisms of brain-derived neuroptrphic factor (BDNF), whose expression is also regulated by cortisol (Shalev et al., 2009), have been associated with AD risk in women (Fukumoto et al., 2010). Lastly, neuroinflammation, an emerging AD pathological feature, has been proposed as a potential mediator of the estrogen depletion effects on cognition (Au et al., 2016).

From a methodological point of view, this study represents a model summary of obstacles encountered when conducting a systematic review and meta-analysis. The heterogeneity pertaining to methodology, outcome assessment, and statistical analysis comprised the most important issue. Specifically, cross-sectional, longitudinal and case-control approaches were used, implementing different adjustment variables and dementia was assessed either via clinical diagnosis or screening instruments. Statistically, most studies treated age at menopause categorically with non-standard categories, whereas dementia risk was estimated by logistic or Cox regression analysis, depending on study design. Sensitivity, subgroup and meta-regression analyses were conducted but were inadequate to resolve heterogeneity. Although findings of the cognitive function systematic review seemed more consistent, the statistical variation did not allow quantitative synthesis.

Recall bias is an inherent limitation; cognitively impaired subjects are more likely to inaccurately recall timing of menopause. Furthermore, all but 2 studies in the dementia analysis included women with natural and surgical menopause jointly, comprising a possible source of bias due to operational indication. Previous studies on surgical menopause have found significant associations with the risk of dementia (Bove et al., 2014; Rocca et al., 2007), a finding consistent with the estrogen deprivation hypothesis. Similarly, all studies in the dementia analysis included participants having used HT, even though most of them adjusted for it. Sensitivity analyses were performed for both confounders without deescalating heterogeneity, whereas many studies assessing cognitive function excluded HT users and women with non-physiological menopause. Other factors affecting lifetime estrogen exposure, like parity, breastfeeding and oral contraceptive use could not be addressed in this review. Lastly, the diverse geographical pattern of age at menopause (Gold, 2011), necessitates the inclusion of Asian and African populations in future literature, as all but 2 Chinese eligible studies concerned Western countries.

The methodological approach based on current guidelines and including an intensive search of published and grey literature is the major strength of this study. More than 12,000 articles were screened for eligibility with no language or publication year restriction. A rigorous contact with authors was carried out to maximize the available evidence. Despite heterogeneity, the conducted meta-analysis included almost 20,000 women and no publication bias was evident. In addition, sensitivity, subgroup, and meta-regression analyses were carried out to evaluate the effect of potential confounders.

5. Conclusions

In conclusion, this systematic review does not seem to support an association between increasing duration of exposure to endogenous gonadal steroids and lower risk of dementia but suggests a potential association with better cognitive performance and delayed cognitive decline. Current literature is methodologically insufficient to overcome the obstacles derived from the long time-gap between exposure and outcome, as well as the interference of many potential confounders, pointing towards current knowledge gaps and future research priorities in this clinically relevant area. Prospective properly designed studies with larger and homogeneous samples and longer follow-up periods, starting closer to menopause and assessing cognition at multiple time-points, should be conducted, considering all intermediate covariates. The conflicting evidence suggesting neuroprotective actions of sex hormones by experimental studies on the one hand and the inefficacy of estrogen-based therapies on humans on the other, necessitates the investigation of other possible mechanisms of female preponderance in dementia.

Conflicts of interest

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.psyneuen.2016.08.003.

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